

Class activity: exploring outliers

Instructions:

- Work individually or with a neighbor to answer the following questions
- To get started, download the [class activity template](#) file
- When you are finished, render the file as an HTML and submit the HTML to Canvas (let me know if you encounter any problems)

Urban housing data

In this activity, we will work with data on housing and urban homeowners. The data are available in the `usdata` package.

Setup:

1. In your *R console*, run the following to install the `usdata` package:

```
install.packages("usdata")
```

2. Now run the following code to import the `urban_owner` data into R.

```
library(tidyverse)
library(usdata)
data("urban_owner")
```

Each row in the `urban_owner` dataset represents one state in the US, plus Puerto Rico and DC. We are interested in the following two variables:

- `poppct_urban`: the proportion of the population living in urban areas
- `pct_owner_occupied`: the proportion of families who own their own home

Questions

1. Run the following code to create a scatterplot of the percent of families owning their own home, vs. the proportion of the population living in urban areas. What do you notice about the scatterplot?

```
urban_owner |>
  ggplot(aes(x = poppct_urban,
             y = pct_owner_occupied)) +
  geom_point() +
  theme_bw() +
  labs(x = "Percent urban population",
       y = "Percent who own their home")
```

2. Fit a linear regression model with `poppct_urban` as the explanatory variable, and `pct_owner_occupied` as the response.
3. Fill in the following code to create a studentized residual plot for your fitted model:

```
data.frame(studentized_resids = rstudent(...),
           yhat = predict(...)) |>
  ggplot(aes(x = yhat, y = studentized_resids)) +
  geom_point() +
  geom_abline(slope = 0, intercept = 0,
             color = "blue", lwd=1.2) +
  geom_abline(slope = 0, intercept = c(-3, -2, 2, 3),
             color = "orange", lty = 2, lwd=1.2) +
  labs(x = "Predicted fraction owning their home",
       y = "Studentized residuals") +
  theme_bw()
```

4. We are typically concerned by observations with studentized residuals > 3 or < -3 . How many such points appear in the plot?